



Course Title: Artificial Intelligence

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Project Title

STUDENT PERFORMANCE PREDICTION SYSTEM

STUDENT PERFORMANCE PREDICTION SYSTEM

Report:

1. INTRODUCTION

In today's education system, evaluating student performance at an early stage is very important. Traditional evaluation methods depend mostly on final examinations, which may not accurately reflect a student's overall performance.

This project, **Student Performance Prediction System**, uses **Machine Learning techniques** to predict students' final grades based on **Attendance, Assignment scores, and Quiz performance**. The system helps teachers identify students who may need additional support and allows institutions to take timely academic decisions.

2. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To analyze student academic data
- To predict final grades using machine learning
- To use attendance, assignment, and quiz scores as input parameters
- To classify students into grades (A–F)
- To provide a user-friendly interface for prediction

3. PROBLEM STATEMENT

Manual evaluation of student performance is time-consuming and may not provide early warnings for weak students. There is a need for an **automated system** that can:

- Predict students' grades before final exams
- Help teachers identify low-performing students
- Improve overall academic performance

4. SCOPE OF THE PROJECT

- Applicable to schools, colleges, and training institutes
- Helps in early performance assessment

- Can be extended to web applications or learning management systems
- Supports academic decision-making

5. DATASET DESCRIPTION

The dataset used in this project contains the following attributes:

Attribute Name	Description
Attendance	Student attendance percentage
Assignment	Average assignment score
Quiz	Quiz performance score
Final_Grade	Final grade obtained by student

The dataset is synthetically generated for demonstration purposes and closely resembles real academic data.

6. METHODOLOGY

The project follows the standard **Machine Learning lifecycle**:

6.1 Data Collection

Student performance data is collected in structured format.

6.2 Data Preprocessing

- Feature selection
- Data scaling using **StandardScaler**
- Splitting data into training and testing sets

6.3 Model Training

Two machine learning models are trained:

- Linear Regression
- Random Forest Regression

6.4 Model Selection

The model with the **highest R^2 score** is selected as the best model.

6.5 Prediction

The trained model predicts final scores and assigns grades.

7. MACHINE LEARNING ALGORITHMS USED

7.1 Linear Regression

A simple supervised learning algorithm used for predicting continuous values.

7.2 Random Forest Regression

An ensemble learning algorithm that:

- Combines multiple decision trees
- Improves accuracy
- Reduces overfitting

Random Forest performed better and was selected as the final model.

8. SYSTEM ARCHITECTURE

Input → Preprocessing → Model Training → Prediction → Grade Classification → Output

Inputs:

- Attendance
- Assignment score
- Quiz score

Outputs:

- Predicted Final Score
- Grade (A–F)

9. IMPLEMENTATION DETAILS

Tools & Technologies Used:

- Python
- Pandas & NumPy (Data Handling)
- Scikit-learn (Machine Learning)
- Matplotlib & Seaborn (Visualization)
- Tkinter (GUI Development)

Evaluation Metrics:

- Mean Squared Error (MSE)
- R^2 Score

10. GRADE CLASSIFICATION SYSTEM

Score Range	Grade
≥ 90	A
80 – 89	B
70 – 79	C
60 – 69	D
< 60	F

11. USER INTERFACE (GUI)

A **Tkinter-based GUI** is developed where users can:

- Enter attendance percentage
- Enter assignment score
- Enter quiz score
- Click “Predict Grade” button
- View predicted score and grade

This makes the system easy to use for teachers and administrators.

12. RESULTS AND DISCUSSION

- Random Forest Regression achieved higher accuracy
- Attendance and assignments strongly affect final grades
- The system successfully predicts student performance
- The model helps identify students needing improvement

13. ADVANTAGES OF THE SYSTEM

- Saves time and effort
- Provides early performance prediction
- Improves academic decision-making
- Easy to use
- Scalable and extendable

14. LIMITATIONS

- Uses a limited dataset
- Prediction accuracy depends on data quality
- Does not include behavioral or psychological factors

15. FUTURE ENHANCEMENTS

- Web-based application using Flask or Django
- Database integration (MySQL)
- Student performance dashboard
- At-risk student alert system
- Inclusion of more academic parameters

16. CONCLUSION

The **Student Performance Prediction System** successfully demonstrates how machine learning can be used in education to predict students' grades using attendance, assignments, and quiz data. The system improves transparency, supports teachers, and helps students perform better academically.

17. REFERENCES

1. Scikit-learn Documentation
2. Python Official Documentation
3. Machine Learning by Tom Mitchell
4. Research papers on Educational Data Mining
